

Hemoglobin Folding Lab 2

1. Hemoglobin is composed of multiple protein chains. Therefore, hemoglobin is said to have a quaternary structure. How many protein chains does normal hemoglobin have (1A3N)? Given this, which term best describes hemoglobin: monomer, dimer, trimer, tetramer, or pentamer?

a. Four, two beta and two alpha. Given this, hemoglobin is a tetramer.

2. How many amino acids does normal hemoglobin (1A3N) have?

a. Hemoglobin has 146 amino acids.

3. Briefly look at the amino acid sequence of each protein chain in normal hemoglobin. Are the various protein chains within normal hemoglobin (1A3N) likely to be identical? Answer Yes/No and provide a brief explanation for your answer.

a. No, all four chains are not identical, but the alpha chains are identical to one another and the beta chains are identical to one another.

4. Based on the nature of their side-chains, what type of interaction do amino acids 85 and 88 in the 2nd protein chain form with each other's side chains? Your choices are 'a covalent bond', 'ionic bond', 'hydrogen bond', or 'hydrophobic interaction'.

a. PHE(85) and LEU(88) are both amino acids with hydrophobic side chains and will therefore have hydrophobic interactions with one another.

5. Do you predict amino acids 85 and 88 will be found on the surface of the protein or buried inside? (Explain your answer)

a. We predict that amino acids 85 and 88 will be buried because they want to minimize their contact with water or aqueous solutions due to their hydrophobic nature.

Structure -> Function -> Disease Questions

6. We will now compare the structure of normal hemoglobin (1A3N) with that of sickle hemoglobin (2HBS) to understand how a single amino acid change in hemoglobin leads to sickle cell anemia.

a. Does the overall tertiary structure differ in an evident way? Describe and explain in your own words

i. **Answer: No, it looks like the same overall tetramer. There are still two alpha and beta chains.**

7. We will now explore the specific alteration in the hemoglobin protein that results in the

development and manifestations of sickle cell anemia. A single amino acid substitution at position.

- a. Identify the protein chains (A, B, C, etc) in sickle hemoglobin (2HBS) with this specific amino acid substitution.
 - i. **Chain B and Chain D's position 6 are both valine, so both beta protein chains in sickle cell hemoglobin have undergone an amino acid substitution.**
- b. Name the amino acid present in normal hemoglobin (1A3N) that is substituted to valine at position 6 in sickle hemoglobin (2HBS). How does this amino acid differ from valine?
 - i. **The amino acid that valine replaces is glutamic acid. Valine has a hydrophobic side chain whereas glutamic acid has a negatively charged side chain.**
- c. Where is the amino acid at position 6 that is substituted to valine in sickle cell anemia located: on the inside or the outside of the normal hemoglobin protein(1A3N)?
 - i. **The glutamic acid on the normal hemoglobin is on the outside of the protein.**

8. In the sickle hemoglobin structure (2HBS) valine 6 in one sickle hemoglobin protein interacts with phenylalanine 85 and leucine 88 in another sickle hemoglobin protein.

- a. Identify which valine 6 and which phenylalanine 85 and leucine 88 interact in this structure (2HBS) by stating the protein chains (A, B, C, etc) in which each of them are found.
 - i. **The valine 6 in the H chain of the second sickle hemoglobin protein interacts with the phenylalanine 85 and leucine 88 in the first sickle hemoglobin protein's B chain.**
- b. What is the most likely type of interaction between valine 6, phenylalanine 85 and leucine 88? Explain your answer.
 - i. **All three amino acids have hydrophobic side chains and therefore would have hydrophobic interactions between them.**
- c. In unaffected, normal patients, the amino acid located at position 6 in either of the two protein chains of normal hemoglobin does not interact with phenylalanine 85 and leucine 88 in another normal hemoglobin protein. Why?
 - i. **The amino acid located at position 6 in the normal hemoglobin protein is glutamic acid which has a negatively charged side chain whereas phenylalanine 85 and leucine 88 have hydrophobic side chains. Because glutamic acid's side chain has different properties from leucine and phenylalanine, they will not interact.**